SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH219

Timed Online Assignment - July Semester 2022

Fundamentals of Statistics and Probability

Thursday, 17 November 2022 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **EIGHT (8)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and Not Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

MTH219 Copyright © 2022 Singapore University of Social Sciences (SUSS)

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1a

In a game, players select weights at random from a box containing a large number of weights of different weights. The weight W, in kg, of a randomly chosen weight has a continuous uniform distribution over the interval [1,7] kg. A weight is chosen at random.

Question 1a(i)

Compute the probability of drawing a weight lighter than 6.5kg.

(3 marks)

Question 1a(ii)

To win a toy, a player must select 4 weights and wins if the weights of the heaviest weight is more than 6.5 kg. Hence, compute the probability that a particular player will win the toy.

(3 marks)

Question 1a(iii)

To win $10, a player must select 7 weights and wins if more than 4 weights are lighter than 4kg. Find the probability of winning and hence calculate how much does the gamemaster need to charge per game to break even.

(6 marks)

Question 1b

A continuous random variable X has distribution $N(\mu, \sigma^2)$. It is known that $P(X < k) = 0.05$ and $P(X < -5) = 0.95$ for some constant $k < -5$.

Question 1b(i)

Compute $P(k \leq X < -5)$ and hence write down the value of $P(\mu < X < -5)$. Hint: $P(X < k) + P(k \leq X < -5) + P(X \geq -5) = 1$.

(3 marks)

Question 1b(ii)

Express μ in terms of k.

(2 marks)

Question 1b(iii)

Assume that $\sigma^2 = 10$ for the rest of the question. Determine the value of μ.

(4 marks)

Question 1b(iv)

Determine the value of k and hence calculate the probability $\cancel{P(-k < X < k)}$ $P(-|k| < X < |k|)$.

(4 marks)

Question 2a

For events A and B, it is given that

$$P(A) = 0.4, \; P(A \cup B) = 0.76, \; P(B|A) = 0.6.$$

Question 2a(i)

Calculate $P(B)$ and show that events A and B are independent.

(5 marks)

Question 2a(ii)

An event C is such that A & C are mutually exclusive and $P(C) = 0.5$.

(i) Sketch a Venn Diagram representing Events A, B and C. Mark out the area representing the probability $P(A' \cup B), P(C \cap B)$ and $P(B' \cup C)$.

(ii) Write down an equation $P(C \cap B)$ and $P(B' \cap C)$.

(iii) Find the minimum value of $P(B' \cap C)$.

(6 marks)

Question 2a(iii)

Can two independent, non-zero probability events A and B be mutually exclusive at the same time? Explain your answer. Non-zero probability means that $P(A) > 0$ and $P(B) > 0$.

(4 marks)

Question 2b

In a certain game, the score X is a random variable with probability distribution given in the following Table Q2(b):

x	-10	-4	5	7	10
$P(X = x)$	0.13	0.42	p	q	0.125

Table Q2(b)

Question 2b(i)

Given that $E(X) = 0.145$, find p and q.

(3 marks)

Question 2b(ii)

Calculate $E((X + 10)^2)$.

(2 marks)

Question 2b(iii)

Calculate $Var(X)$ and $SD(X - 8)$.

(3 marks)

Question 2b(iv)

The gamemaster wants to change the scores such that the game become fair i.e. $E(X) = 0$. Propose a simple way for him to achieve this.

(2 marks)

Question 3a

Fish is delivered daily to a Sushi restaurant. From the delivery truck, the masses of Atlantic mackerels have the distribution $N(3400, 350^2)$ while the masses of Atlantic salmon have the distribution $N(\mu, \sigma^2)$, both in grams.

Question 3a(i)

Let X be the ~~average~~ total mass of 5 mackerels and 3 salmons. Given that $P(X < 29084.5) = P(X > 31915.5) = 0.1$, determine μ and σ.

(6 marks)

Question 3a(ii)

Apply the normal distribution model to find the probability that the total mass of 5 mackerels differs from 4 times the mass of a salmon by at most 800g.

(5 marks)

Question 3a(iii)

Atlantic mackerels need to be prepared before they can be used in the restaurants. This process reduces the mass of the mackerels by 40%. Calculate the probability that the mass of 10 randomly chosen mackerel, after preparation, drops below 19.5 kg.

(4 marks)

Question 3a(iv)

Mackerels which are above 4kg are considered large. A mackerel was chosen at random. Given that the mackerel is large, find the probability that the mackerel has mass greater than 4.3kg.

(5 marks)

Question 3a(v)

Half of the large mackerels have mass greater than Y grams. Calculate the value of Y.

(5 marks)

Question 4a

There are 12 cards on the table, comprising of 6 "Kings", 4 "Queens" and 2 "Aces". The cards are face down (i.e. you can only see the card back) and a player has no way of determining the card unless he turns it over. A player randomly turns over a card, one at a time, until he has a pair of cards which are of the same type. Let the random variable X denote the total number of cards the player turned over.

Question 4a(i)

Determine the probability distribution (PMF) of X. Hint: X is unable to exceed 4 because there are only 3 types of cards, thus the 4th card will surely make a pair with one of the previous 3.

X= **number of cards turned over**	2	3	4
$P(X = x)$			

<u>Table Q4(a)(i)</u>

(6 marks)

Question 4a(ii)

Calculate $E(X)$ and $Var(X)$.

(6 marks)

Question 4b

A Ramen shop only sells 2 types of Ramen – Shio Ramen and Shoyu Ramen. (Shio means Salt, Shoyu means Soy Sauce). The number of bowls of Shio Ramen sold may be assumed to be a Poisson random variable with mean 20 bowls every hour, while the number of bowls of Shoyu Ramen sold may also be assumed to be a Poisson random variable, but with mean 17 bowls every half-hour. The number of Shio and Shoyu Ramen sold are independent of each other.

Question 4b(i)

Applying the Poisson distribution, compute the probability that in a given 10 minutes period,

(i) exactly 3 Shio Ramen was sold,
(ii) exactly 5 Shoyu Ramen was sold,
(iii) exactly 3 Shio Ramen AND 5 Shoyu Ramen was sold.

(7 marks)

Question 4b(ii)

Calculate the probability that in a given 15 minutes period, more than 17 bowls of ramen are sold in total.

(6 marks)

----- END OF PAPER -----